Online Voting System Technical White Paper
Version 1 – Extended Draft Design Document

This extended draft is a pre-implementation technical design intended to evolve into Version 2 after completion of the mobile application, testing, and external review.

Executive Summary

This project proposes a secure online voting platform for South African elections with separate voter and administrator interfaces, emphasizing confidentiality, integrity, availability, transparency, and accessibility. This project proposes a secure online voting platform for South African elections with separate voter and administrator interfaces, emphasizing confidentiality, integrity, availability, transparency, and accessibility. This project proposes a secure online voting platform for South African elections with separate voter and administrator interfaces, emphasizing confidentiality, integrity, availability, transparency, and accessibility.

Background

Traditional elections face logistical challenges, long queues, costs, accessibility barriers, and slow reporting. Online voting must address these while preserving trust. Traditional elections face logistical challenges, long queues, costs, accessibility barriers, and slow reporting. Online voting must address these while preserving trust. Traditional elections face logistical challenges, long queues, costs, accessibility barriers, and slow reporting. Online voting must address these while preserving trust. Traditional elections face logistical challenges, long queues, costs, accessibility barriers, and slow reporting. Online voting must address these while preserving trust.

System Goals

Design principles include usability, security-by-design, resilience, auditability, privacy, scalability, inclusivity, and compliance. Design principles include usability, security-by-design, resilience, auditability, privacy, scalability, inclusivity, and compliance. Design principles include usability, security-by-design, resilience, auditability, privacy, scalability, inclusivity, and compliance. Design principles include usability, security-by-design, resilience, auditability, privacy, scalability, inclusivity, and compliance.

Architecture

Frontend web administration, mobile voter app, backend API, MongoDB database, authentication, logging, monitoring, and future cloud deployment. Frontend web administration, mobile voter app, backend API, MongoDB database, authentication, logging, monitoring, and future cloud deployment. Frontend web administration, mobile voter app, backend API, MongoDB database, authentication, logging, monitoring, and future cloud deployment. Frontend web administration, mobile voter app, backend API, MongoDB database, authentication, logging, monitoring, and future cloud deployment. Frontend web administration, mobile voter app, backend API, MongoDB database, authentication, logging, monitoring, and future cloud deployment.

Authentication

Role separation, secure passwords, MFA roadmap, session management, device verification, and future national identity integration. Role separation, secure passwords, MFA roadmap, session management, device verification, and future national identity integration. Role separation, secure passwords, MFA roadmap, session management, device verification, and future national identity integration. Role separation, secure passwords, MFA roadmap, session management, device verification, and future national identity integration. Role separation, secure passwords, MFA roadmap, session management, device verification, and future national identity integration.

Vote Lifecycle

Registration, authentication, ballot issuance, encrypted transmission, storage, counting, verification, and publication. Registration, authentication, ballot issuance, encrypted transmission, storage, counting, verification, and publication. Registration, authentication, ballot issuance, encrypted transmission, storage, counting, verification, and publication. Registration, authentication, ballot issuance, encrypted transmission, storage, counting, verification, and publication. Registration, authentication, ballot issuance, encrypted transmission, storage, counting, verification, and publication.

Threat Model

Credential theft, insider abuse, malware, MITM, denial-of-service, database compromise, social engineering, supply-chain attacks. Mitigations accompany every threat. Credential theft, insider abuse, malware, MITM, denial-of-service, database compromise, social engineering, supply-chain attacks. Mitigations accompany every threat. Credential theft, insider abuse, malware, MITM, denial-of-service, database compromise, social engineering, supply-chain attacks. Mitigations accompany every threat. Credential theft, insider abuse, malware, MITM, denial-of-service, database compromise, social engineering, supply-chain attacks. Mitigations accompany every threat. Credential theft, insider abuse, malware, MITM, denial-of-service, database compromise, social engineering, supply-chain attacks. Mitigations accompany every threat.

Security Controls

TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap. TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap. TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap. TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap. TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap. TLS, password hashing, RBAC, encryption, backups, immutable logs, monitoring, least privilege, secure coding, code review, penetration testing roadmap.

Fraud Prevention

One-person-one-vote validation, duplicate detection, anomaly monitoring, audit trails, administrative separation of duties. One-person-one-vote validation, duplicate detection, anomaly monitoring, audit trails, administrative separation of duties. One-person-one-vote validation, duplicate detection, anomaly monitoring, audit trails, administrative separation of duties. One-person-one-vote validation, duplicate detection, anomaly monitoring, audit trails, administrative separation of duties. One-person-one-vote validation, duplicate detection, anomaly monitoring, audit trails, administrative separation of duties.

Privacy

Identity separated from ballots, minimal personal data, secure retention, lawful deletion where applicable. Identity separated from ballots, minimal personal data, secure retention, lawful deletion where applicable. Identity separated from ballots, minimal personal data, secure retention, lawful deletion where applicable. Identity separated from ballots, minimal personal data, secure retention, lawful deletion where applicable.

Testing Strategy

Unit, integration, security, usability, load, recovery, penetration, and user acceptance testing. Unit, integration, security, usability, load, recovery, penetration, and user acceptance testing. Unit, integration, security, usability, load, recovery, penetration, and user acceptance testing. Unit, integration, security, usability, load, recovery, penetration, and user acceptance testing.

Current Progress

The web platform has been developed. The mobile voter application is in progress. Future work includes independent review and production hardening. The web platform has been developed. The mobile voter application is in progress. Future work includes independent review and production hardening. The web platform has been developed. The mobile voter application is in progress. Future work includes independent review and production hardening. The web platform has been developed. The mobile voter application is in progress. Future work includes independent review and production hardening.

Roadmap

Complete mobile application, security review, documentation updates, researcher feedback, IEC engagement, Version 2 publication. Complete mobile application, security review, documentation updates, researcher feedback, IEC engagement, Version 2 publication. Complete mobile application, security review, documentation updates, researcher feedback, IEC engagement, Version 2 publication. Complete mobile application, security review, documentation updates, researcher feedback, IEC engagement, Version 2 publication.

Conclusion

This Version 1 is a design document rather than a certification of security. It captures engineering intent and will be refined through implementation and expert review. This Version 1 is a design document rather than a certification of security. It captures engineering intent and will be refined through implementation and expert review. This Version 1 is a design document rather than a certification of security. It captures engineering intent and will be refined through implementation and expert review.